

# PySpark Data Transformation Queries

```
spark.read.csv(..., header=True, inferSchema=True)
df.printSchema()
df.withColumnRenamed('old', 'new')
df.dropDuplicates()
df.fillna({'Sales_Amount':0})
df.withColumn('Sale_Date', col('Sale_Date').cast('date'))
df.filter(col('Sales_Amount')>1000)
sales.join(products, 'Product_ID', 'inner')
df.groupBy('Territory').agg(sum('Sales_Amount').alias('Revenue'))
df.orderBy(col('Sales_Amount').desc())
Window.partitionBy('Territory').orderBy(col('Sales_Amount').desc())
row_number().over(windowSpec)
rank().over(windowSpec)
dense_rank().over(windowSpec)
df.write.format('delta').mode('overwrite').saveAsTable('gold.sales_analytics')
spark.sql('OPTIMIZE gold.sales_analytics')
spark.sql('VACUUM gold.sales_analytics RETAIN 168 HOURS')
```